① $S_A = \sum_{n=1}^{\infty} \frac{n^2}{2^n}$

$a_n = \frac{n^2}{2^n}$

$a_{n+1} = \frac{(n+1)^2}{2^{n+1}}$

$\Rightarrow$ Ratio test $= \frac{a_{n+1}}{a_n} = \frac{(n+1)^2}{(2^{n+1})} \frac{(2^n)}{(n^2)(2)}$

$L \Rightarrow \lim_{n\to\infty} \left(\frac{n+1}{n}\right)^2 \frac{1}{2}$

$\Rightarrow \lim_{n\to\infty} \frac{n+1}{n} = 1$

$\boxed{L = \frac{1}{2}} < 1 \longrightarrow$ Thus the series S_A Converges.

② $S_B = 1 + \frac{1}{2} + \frac{1}{8} + \frac{1}{16} + \frac{1}{64} + \frac{1}{128} + \frac{1}{512} - -$

$= \frac{1}{2^0} + \frac{1}{2^1} + \frac{1}{2^3} + \frac{1}{2^4} + \frac{1}{2^6} + \frac{1}{2^7} + \frac{1}{2^9} -$

Sum of two series

$\frac{1}{2^0} + \frac{1}{2^3} + \frac{1}{2^6} + \frac{1}{2^9} \rightarrow r_1 = \frac{1}{2^3}$

$\frac{1}{2^1} + \frac{1}{2^4} + \frac{1}{2^7} - -$

$r_2 = \frac{1}{2^3}$

Both are converging series